



Optical networks management and control: A review and recent challenges

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ABSTRACT

In the last twenty years, optical networks have witnessed recurrent changes in their management and control architecture. In this paper, we present a historical timeline and a future perspective of the evolution of optical network management and control deployed for Wavelength Switched Optical Networks (WSON), Elastic Optical Networks (EON) and (multilayer) Data Center Networks.

Early implementations of WSON envisaged a static and centralized provisioning approach supported by the Management Plane only. Gradually, the requirement of accommodating more network dynamics in WSON, and later in EON, pushed the adoption of a distributed control, mostly supported by vendor-dependent implementations of the Generalized MultiProtocol Label Switching (GMPLS) protocol suite. The drawbacks of the fully distributed GMPLS-based control, such as resource contention, suboptimal resource usage, and complex computations (e.g., to account for physical layer constraints) showed the necessity to bring back some of the routing/provisioning functions to a centralized Path Computation Element (PCE) capable of accounting for e.g. physical impairments and interworking with GMPLS.

The centralized control then gained its momentum and brought a radical change in network control, through the separation of data and control plane introduced by the paradigm of Software Defined Networking (SDN). Such an approach has been gradually extended to optical network control.

The paper, eventually, presents the most advanced control techniques, namely the intent-based networking, the observe/decide/act state-based approach providing for autonomic optical network and the (closed-loop) zero-touch service management approach. Advanced traffic conditioning techniques are also detailed, namely the in-band telemetry and the exploitation of Programming Protocol-Independent Packet Processors (P4) language capabilities as well as solutions tailored for data center networks: all of them are still in a research stage and to be integrated within future optical network architectures.

1. Introduction

The evolution of optical networks has seen its first milestone in switching capability when the wavelength became the granularity used to set-up and route the communication between node pairs using lightpaths (wavelength paths) [1]. Based on the definition of four “eras” in optical networking in Ref. [2] this paper expands them focusing on the corresponding management and control architectures:

- (1) the era of *regeneration* corresponds to the time-frame when Wavelength Switched Optical Networks (WSONs) started to be deployed. In WSON the traffic is accommodated by setting up and tearing down lightpaths between a source and a destination possibly in an automatic and fast way. The limited number of wavelengths per link or the node architecture or the path length limitation are typical constraints of the problem to determine both the routes over which these lightpaths should be established and the wavelengths that should be assigned to them.

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Regenerators in the intermediate nodes alleviate the limitation of the connection reach.

- (2) the era of *amplified dispersion-managed systems* corresponds to the introduction of amplification stages in optical networks. Initially, such impairments were managed through a manual management approach, later the network dynamicity called for some automatic “calibration” through impairment models or direct network probing. This spurred the accounting of physical impairments at the control plane level [3].
- (3) the era of *amplified coherent systems* was opened thanks to the use of advanced (multilevel) modulation formats and the coherent transmission and detection in the optical domain. The main notable impact of these new techniques was the advent of elastic optical networks (EON) with improved spectral efficiency and better matching between allocated optical network resources and traffic demands. Moreover, the use of coherent receivers enabled end-to-end monitoring (e.g., of bit error rate – BER – or optical signal to noise ratio – OSNR) through digital signal processing, offering more chances for network and services observation [4, 5].
- (4) the era of *high capacity networking* is the current era where optical networks are searching novel dimensions to increase the transport capacity. Among current research topics we mention the usage of bands beyond C for transmission (e.g. E, S, and L) [6,7] or advanced domains such as the orbital angular momentum (OAM) multiplexing [8] and space division multiplexing [9,10].

The paper is organized as follows: the introduction continues by reviewing the literature on network management/control definitions and considered optical network architectures. In Sec. 2 we present a review of distributed optical network control (GMPLS, PCE and their interworking), Sec. 3 is devoted to Software Defined Networking applied to optical networks, while Sec. 4 deals with multi-layer network control. Finally, Sec. 5 and 6 cover advanced network control techniques and recent challenges which are still at the research phase.

In Sec. 7 conclusions are then drawn.

1.1. Management vs. control

From a strictly functional viewpoint, the management and control in networking refer to a different set of functions (i.e., planes) defined as follows.

Management Plane: it is involved with the planning, installation, configuration, provisioning and supervision of the network infrastructure in order to allow a coordinated operation of the network elements and assure the network functions as efficiently and smoothly as possible and in accordance to user requirements. Network management includes a set of management activities and processes with specific goals and actions, not necessarily disjoint to each other, i.e., Fault, Configuration, Accounting, Performance and Security (FCAPS) [11,12]. Different levels of management can be identified (i.e., layers) according to a hierarchy of abstractions and management areas: business, service, network, element management layers. Over time, management operations have been typically deployed according to a centralized approach in a management system at the different layers and relied on manual configurations by human operators. However, recently, automation has been embraced along with advanced techniques (e.g., machine learning) to improve efficiency.

Control Plane: it is involved with learning and updating the network topology and deciding what to do with incoming packets or connection requests node-by-node from source to destination. Control plane functions, such as participating in routing protocols, run in an architectural control element. In (virtual) connection-oriented networks, the control plane is concerned with all aspects of configuring connections through topology discovery, routing and signaling. Control plane operations are intrinsically automated since they leverage the protocols among nodes

(e.g., to identify paths) and the autonomous decisions of nodes (e.g., use and selection of back-up paths) using the acquired information (e.g., topology, monitoring).

Either or both above planes are usually coupled with the transport (or data) plane which carries the user data and comprises network equipment, such as traffic cards, optical line interface cards, switch fabrics, backplanes, and fiber plant.

Early optical networks exhibited a highly static behavior and were managed by centralized management systems at different layers. This behavior could be modeled as the interaction of a transport plane with a management plane. Indeed, low connection provisioning rate was the norm of operation and it was possible for a management plane to keep track of detailed connection information and configuration commands for connection set-up (even manually) without excessive processing burden.

With the demand of fast and dynamic connection provisioning, dynamic optical networks started to be deployed with a control plane interworking with the management and transport planes. The control plane increases automation by moving some of the network intelligence from the management system down to the network elements, such as resource and service discovery. As a result, the network elements have access to complete network topology and resource information, and can use it to autonomously establish and route optical connections while optimizing resource usage, i.e., switched connections. Switched connections introduce a radically different approach for connection provisioning: following a signaled client request, the control plane autonomously takes care of all the path computation and the signaling which sets up the connection. Therefore, the management plane may not be directly involved during the connection set-up phase.

In the optical transport field, the control plane has been called differently in the different standardization bodies. In the Internet Engineering Task Force

(IETF), the optical control plane is referred to as Generalized Multi-Protocol Label Switching, or GMPLS [13]. Within the International Telecommunication Union - Telecommunication (ITU-T), the optical control plane initiative is referred to as Automatic Switched Optical Network (ASON) [14]. More specifically, the ITU-T has defined the architecture and requirements for an optical control plane, whereas the IETF has mostly focused on developing the control plane protocols. With the advent of Software Defined Networking (SDN), the control plane and the data plane were totally decoupled, giving rise to a new perspective in optical network control architectures.

1.2. Wavelength switched optical networks (WSO)

The first example of optical network switching occurred when the wavelength multiplexing technique was adopted for various data traffic flows (IP, SONET-SDH, Ethernet, etc.) at the optical layer. Originally, the provisioning problem in WDM networks was solved at the management plane level by an operator-driven centralized approach. Subsequently, the introduction of the optical control plane allowed the simplification and automation of provisioning in WDM networks. The well-known GMPLS protocol suite has been for a long time the dominant implementation of the control plane developed by the IETF to facilitate the establishment of Generalized Label Switched Paths (G-LSPs) [13], involving signaling, routing, and resource management functions, and link management (e.g., connectivity verification). GMPLS implements all functional entities necessary for controlling an ASON, even beyond the pure optical domain. Indeed, GMPLS is capable of setting up LSPs in a variety of data plane technologies, defining the first generation of WSON.

Ethernet interfaces have then rapidly become the main transmission standard for customer equipment in WSON. In addition to the already standardized interfaces at 1, 10, 40 and 100 Gb/s Ethernet, the IEEE is defining the requirements for a 400 Gb/s interface for the backbone (IEEE P802.3bs 400 Gb/s Ethernet Task Force). Still, traditional WSONs

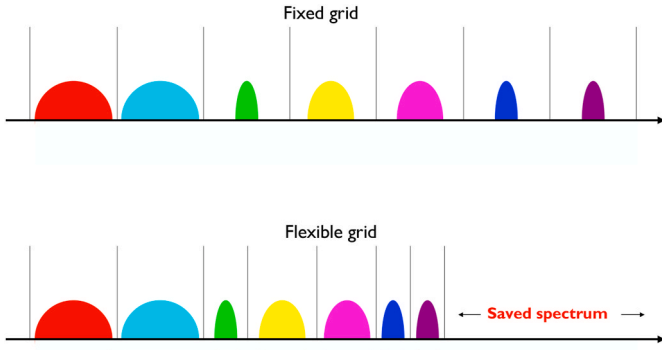


Fig. 1. Examples of fixed- and flexible-grid.

typically operate on a fixed wavelength ITU-T grid, but the most used channel spacing at 50 GHz imposes unnecessary limitations for future transmission rates exceeding 100 Gb/s, when using more complex modulation formats (such as 16QAM) [15]. To allow wavelength multiplexing of optical channels with spectra wider than 50 GHz, ITU-T has introduced the so-called flexible DWDM grid, standardized in G.694.1, which assigns to each channel a slot identified by a central frequency, with spectral width multiple of 12.5 GHz.

1.3. Elastic optical networks

The need of high spectral efficiency to increase fiber utilization and the availability of channels with different bandwidth requirements have been the main drivers for Elastic Optical Networks (EONs).

EONs are presented in Ref. [16]. The authors investigated the topic, considering the network architecture (different node architectures with a performance comparison in terms of scalability and flexibility), the routing and spectrum assignment (RSA) approaches (focusing on spectrum fragmentation and presenting the conventional spectrum allocation policies, including the mathematical modeling and the theoretical computational complexity for different problems).

On one hand, coherent technology has enabled the support of several modulation formats meeting the best trade off between spectral efficiency and optical reach: thus, a given requested bit rate (e.g., 200 Gb/s) can be supported, depending on the path and the physical impairments, by a proper format (e.g., PM-QPSK or PM-16QAM) requiring a specific bandwidth. On the other hand, the traffic rate requested by the per source-destination pairs may be heterogeneous (e.g., 100 and 200 Gb/s channels), thus characterized by different bandwidth requirements. The adoption of a fixed grid in such a scenario should drive the design of the channel spacing according to the largest signal bandwidth. Thus, the presence of channels with a variety of bandwidth requirements makes the classical ITU-T fixed grid inefficient. A more effective use of the spectrum is preferred. To this end, ITU-T introduced the flexible grid [17], for which switched channels are characterized by a central frequency and a configurable *slot width*. In particular, channels may assume

a nominal central frequency $f = 193.1 \text{ THz} + n \times 0.00625 \text{ THz}$, where n is a positive or negative integer including 0; the slot width (equal or higher than the required bandwidth) indicates the amount of reserved optical spectrum and it is defined to be $m \times 12.5 \text{ GHz}$, where m is an integer greater than or equal to 1. Fig. 1 shows an illustrative example of channels with different bandwidth requirements in the presence of fixed and flexible grids.

The adoption of flexible grid and EONs is supported by the enabling technology. Firstly, while classical Wavelength Selective Switches (WSSs) only supported fixed channel spacing, the introduction of bandwidth-variable WSSs (BV-WSSs) [18] permitted to filter (thus, to switch) configurable portions of the spectrum, finally realizing the flexible grid. Secondly, the other essential technology is the bandwidth variable transponder [19]: i.e., a transponder generating (and detecting) signals with different bandwidth requirements due to the support of multiple rates, modulation formats, and code rates. This elasticity – or, in general, flexibility – in terms of bandwidth, modulation format, rates, code rate, and channel spacing has made it possible to realize EONs. The use of flexible grid has been evaluated to save around 30% of fiber spectrum [20].

1.4. Data center networks

Data center networks represent one of the fastest growing network segments, with a forecast cumulative annual growth rate approaching 24.7%, exceeding 20 ZB/year in 2021 according to Ref. [21], with more than 70% of the traffic being exchanged within the data center itself. To handle such traffic, a wide range of intra-data center architectures has been devised, based on electrical switching elements [22]. However, currently used electronic interconnection networks are approaching their limits in terms of bandwidth, wiring and power density, and energy consumption. The introduction of optics can help to face these issues, allowing the support of higher capacities and the reduction of power consumption and footprint. Interconnectivity can be guaranteed using different optical network architectures. To reach the huge capabilities offered by optics in a flexible way, it is necessary that effective controller and schedulers are implemented with guarantees of low packet latency and high throughput. The scheduling problem in generic switching architectures has been widely investigated and many proposals have been made, addressing the challenges and potentialities of packet-based network networks for data centers.

1.5. Physical layer issues

An important aspect in optical networks is related to the physical impairments which cumulate along the path and degrade the lightpath quality of transmission (QoT) [23,24], finally limiting the optical reach. Physical impairments due to signal propagation in fiber can be classified as linear (e.g., attenuation, chromatic dispersion), if they do not depend on signal power, and non-linear (e.g., self-phase modulation (SPM), cross-phase modulation (XPM)), if they depend on signal power.

Table 1
Physical layer impairments.

Linear effects	Non-linear effects	Other effects
Fiber attenuation: optical signal power is attenuated during signal propagation in fiber	Self-phase modulation (SPM): because of the Kerr effect, the power dependence of the fiber refractive index causes a phase shift induced by the power of the signal itself	Amplified spontaneous Emission (ASE): spontaneous emissions add noise to the signal during its amplification
Chromatic dispersion (CD): Different spectral components of the pulse travel at different group velocities in fiber causing a pulse broadening.	Cross-phase modulation (XPM): because of the Kerr effect, the power dependence of the fiber refractive index causes a phase shift in the signal due to other signals propagating in the fiber at different carrier frequencies.	Crosstalk: the power leak among OXC ports causes an interference among signals
Polarization mode dispersion (PMD): two different polarization of the signal travel at different velocities in fiber, due to imperfections and asymmetries, causing a pulse broadening		

Table 2
Examples of margins assumed within QoT estimation [24].

Effect	Value
Polarization dependent loss	1.5 dB
Filtering	1 dB
Crosstalk	0.5 dB
Ageing	2 dB
XPM and other non-linear effects	1 dB
Total	6 dB

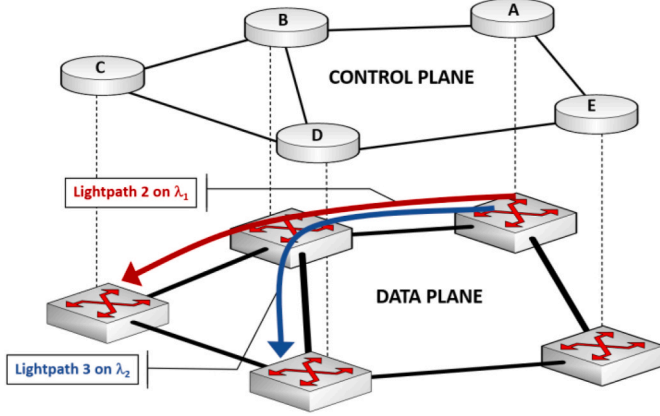


Fig. 2. GMPLS-controlled optical network.

Moreover, other effects – such as amplified spontaneous emission – have to be considered. Table 1 summarizes the main optical layer physical impairments.

Optical networks used to be planned in the past according to the longest paths (or the most impaired one). More recently, in order to implement a more dynamic network, lightpath QoT was assumed to be dynamically evaluated and guaranteed for each lightpath request [23, 24]. QoT models are then adopted to evaluate the actual feasibility of a lightpath [24,25]. Such models typically evaluate the accumulation over a path of specific physical impairments, such as amplified spontaneous emission, while accounting others in the worst-case scenario. The latter is the case, for example, of traffic-variant impairments, such as XPM. Indeed, an operator typically assumes XPM in the worst-case scenario (i.

e., all channels lit on); in this way if a lightpath is feasible in the worst-case scenario, it is always feasible with any configuration of the spectrum. If XPM were evaluated based on the actual traffic, the Quality of Transmission (QoT) of already working lightpaths should be re-evaluated each time a new lightpath is setup, which can be an unfeasible approach. Table 2 shows an example of margins (overall of 6 dB) as reported in Ref. [24] for a 10-Gb/s WSON.

Margins may vary with the network, with the operator policies, and with the transmission technology. Finally, margins are then typically removed from the estimated OSNR along a path resulting in a pessimistic but conservative QoT evaluation.

In the literature, several extensions were proposed to exchange physical layer information within the GMPLS protocol suite to enable a distributed QoT-aware setup: more specifically, extensions to the routing or to the signaling protocol [23,24] were proposed. However, this approach might require complex computation engines distributed in the control plane nodes, thus the solution of a PCE (and later of an SDN controller) also responsible for QoT evaluation seemed to relax the complexity of the control plane.

2. Distributed optical network control

In both WSONs and EONs the control plane has to be implemented on a completely separate network. This is a native requirement arising from the huge transmission rate supported by the optical links. Indeed, in case of failures on the data plane there is the need to have a fully operational control plane to drive the recovery procedure. On the other hand, in case of failures on the control plane the data plane should continue its operation.

2.1. Generalized multiprotocol label switching (GMPLS)

Within the GMPLS framework each node is typically associated to a control plane *agent*. However, a single agent can also control multiple nodes. In turn, the agents are connected using a separate network (i.e., the control plane network) that is typically assumed with the same topology of the data plane, but that in general may be completely different. All the messages belonging to control plane protocols are exchanged on the control plane network.

Fig. 2 shows the control and the data planes of a GMPLS-controlled optical network. In this case the control plane topology is identical to

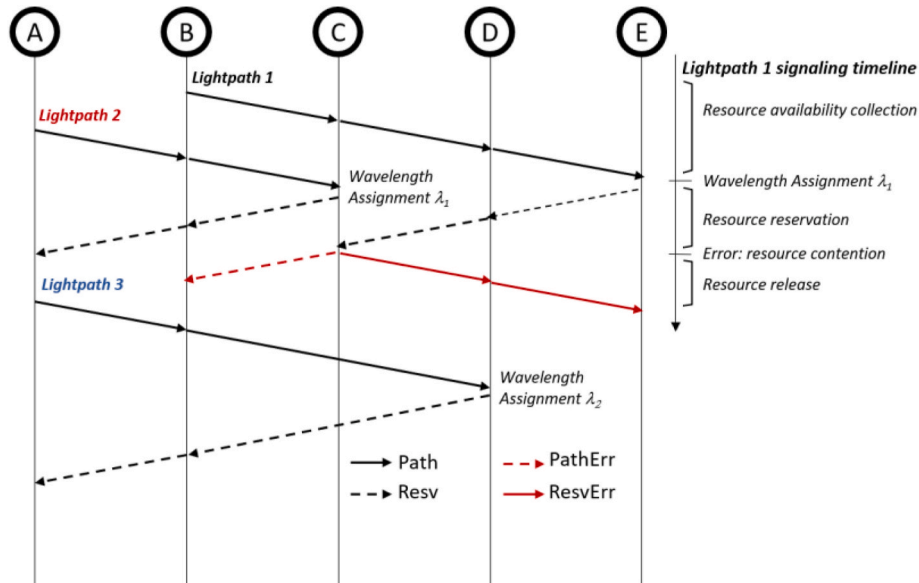


Fig. 3. Distributed control plane reference scheme. GMPLS messages are exchanged on the control plane. The figure illustrated also two established lightpaths in the data plane.

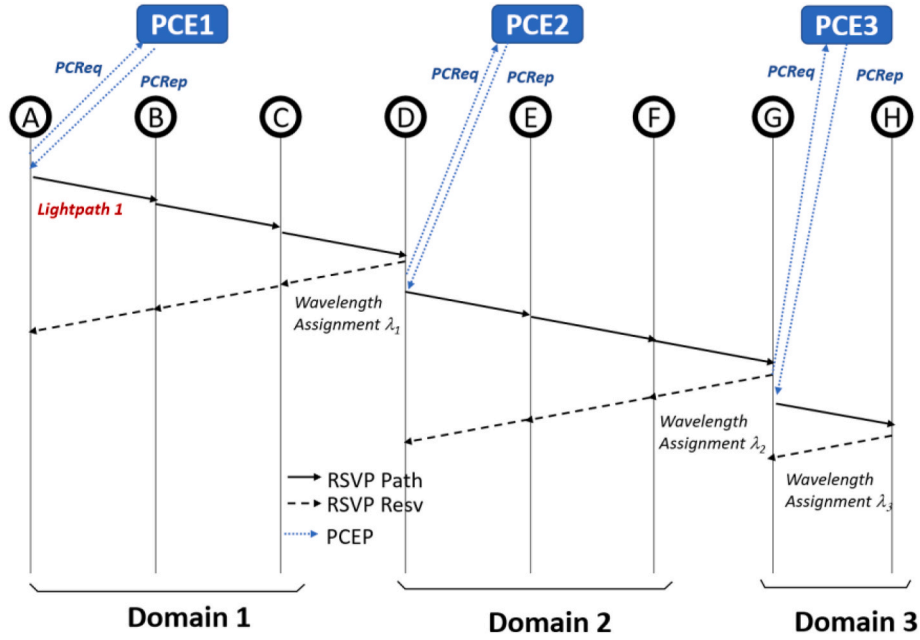


Fig. 4. PCE-based per-domain path computation and signaling.

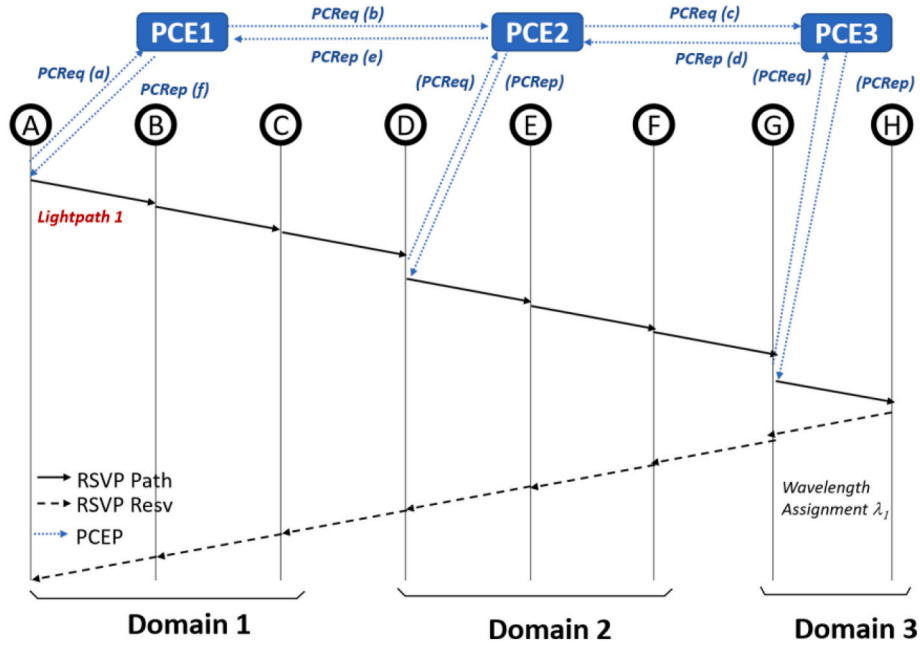


Fig. 5. Inter-PCE backward path computation.

the data plane topology, and each node is provided with its own control agent. The figure also shows two lightpaths established on different wavelengths (i.e., lightpath 2 using λ_1 and lightpath 3 using λ_2).

GMPLS is composed of three main protocols. The routing protocol (i.e., Open Shortest Path First with Traffic Engineering, OSPF-TE) advertises network topology and traffic engineering information (e.g., frequency slot availability in EONs). The signaling protocol (i.e., Resource Reservation Protocol with Traffic Engineering, RSVP-TE) is dedicated to network resource reservation.

The Link Management Protocol (LMP) performs the monitoring of point-to-point links. The GMPLS protocols have been standardized within IETF, with a significant contribution from research work, especially for OSPF-TE and RSVP-TE.

Depending on the network state information advertised by OSPF-TE

each control plane agent can apply effective routing and wavelength/spectrum assignment (RWA/RSA) algorithms [26]. Many RWA/RSA schemes have been proposed using the information advertised by OSPF-TE. Several research works, e.g. Refs. [27,28], focused on determining the required level of details of the information to be advertised (e.g., whether it is necessary to advertise availability of each wavelength for every link or it is sufficient to advertise the number of available wavelengths per link). More advanced work focused on the information needed to be advertised for implementing the distributed computation of backup (e.g., using sparse shared protection) [29–31].

The RSVP-TE protocol is responsible for distributed lightpath establishment. The signaling procedure is initiated by the source agent after successful path computation. RSVP-TE is based on two types of messages: the *Path* message is sent from the source towards the

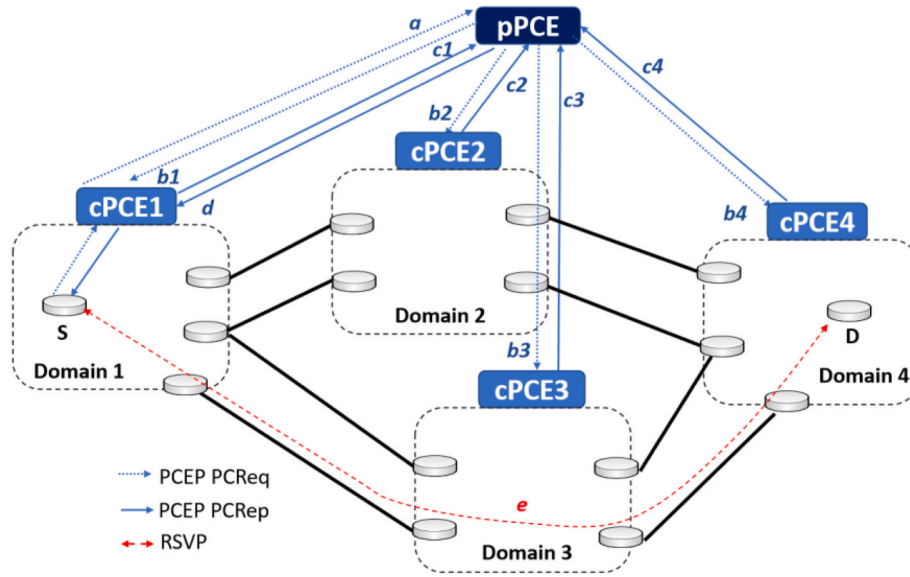


Fig. 6. Hierarchical PCE path computation.

destination (forward direction) and the *Resv* message is sent from the destination toward the source (backward di-rection). The *Path* message collects updated resource availability information (e.g., the list of wavelengths available on each link traversed by the lightpath). This way, the destination agent can perform the wavelength/spectrum assignment relying on collected information. The *Resv* message brings the selected assignment back to the source agent, following the computed path in the backward direction. Every agent traversed by a *Resv* message reserves the selected resources. When the *Resv* reaches the source agent, the transmission can take place on the data plane.

Fig. 3 illustrates the RSVP-TE signaling messages exchanged to establish the two lightpaths depicted also in Fig. 2. Moreover, the figure reports the signaling attempt to establish lightpath 1 that is blocked due to a resource contention in the backward direction. Indeed, both lightpath 1 and lightpath 2 selected the same wavelength, i.e., λ_1 . Thus, only lightpath 2 is successfully established.

The aforementioned signaling procedure may fail in two different ways. The *forward blocking* happens when the RSVP-TE signaling fails in the forward direction. This typically occurs at high network loads due to resource exhaustion and/or incompleteness or inconsistency of the network state information advertised by OSPF-TE. The *backward blocking*, theoretically modeled in Ref. [32], is due to resource contentions and typically occurs with very dynamic traffic conditions such as during restoration. Several works proposed extensions to the RSVP-TE protocol to reduce the backward blocking. The most obvious solution is enabling multiple subsequent signaling attempts [33], however the increased signaling time required for multiple attempts is not acceptable in recovery scenarios. Therefore, other solutions have been proposed including Path message extensions [34]. Similar extensions can be also used to optimize the utilization of sparse wavelength-conversion [35], and collect information regarding optical physical impairments [24].

2.2. The Path Computation Element (PCE)

In lightpath provisioning, the path computation represents the crucial step to achieve effective network resource utilization. As previously explained, using the distributed GMPLS control plane the path computation is performed at the source node, while resources are reserved during the signaling phase. However, distributed path computation may require heavy processing at each control plane node, especially when based on multiple constraints. Moreover, in dynamic traffic conditions it may lead to sub-optimal solutions due to possible

network information inconsistency (e.g., backward blocking) [36]. Thus, the IETF proposed a set of techniques defined under the umbrella of the Path Computation Element (PCE) Architecture. Such techniques rely on path computation performed by dedicated network entities (i.e., the PCEs), that are part of the control plane [37].

The PCE collects network state information (e.g., listening to the OSPF-TE messages) and performs path computation on behalf of network nodes; moreover the PCE is allowed to retrieve network information through direct communication with the Network Management System. Communication between node agents and the PCE is achieved using the Path Computation Element communication Protocol (PCEP) [38].

Besides a more effective and efficient path computation (i.e., the PCE may implement advanced RWA/RSA solutions considering multiple constraints), the utilization of a centralized PCE leads to other important benefits. Indeed, having a single network element where the RWA is performed for each requested lightpath may solve the problem of resource contentions if proper procedures are defined for both the updating of the network state information at the PCE, and the path computation procedure. As an example, the work in Ref. [39] proposes a proactive mechanism to update the PCE database instead of waiting for the OSPF-TE advertisements, thus practically canceling the backward blocking probability. Several experimental works demonstrated the feasibility of the PCE architecture considering also physical impairments during the path computation [40].

On the other hand, the utilization of a PCE may increase the time required to activate a new path (or recover disrupted paths). Indeed, the two-way PCEP communication consumes time, and the centralization of path computations may generate queuing problems. The work in Ref. [41] demonstrates that this issue is negligible in a provisioning scenario, while (depending also on the time required for a single path computation) it could be detrimental during restoration.

2.3. Control of multidomain optical networks via PCE and GMPLS

Due to scalability problems, multi-domain routing protocols (e.g., Border Gateway Protocol, BGP) exchange only reachability information without detailing the resource availability. This way, effective path computation strategies cannot be applied for inter-domain LSPs. Thus, the PCE architecture is also defining path computation procedures with limited resource advertisement to be used in multi-domain and multi-operator networks [42].

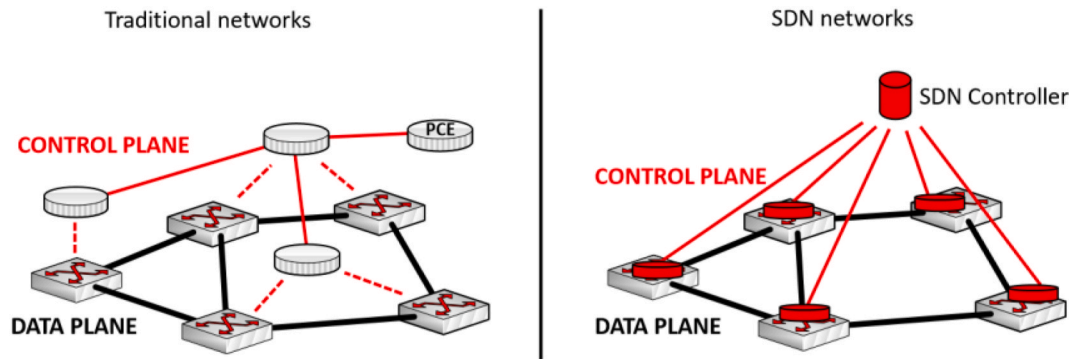


Fig. 7. Control and data plane reference architectures: (a) distributed GMPLS control plane with PCE (control plane topology can be different with respect to data plane topology); (b) SDN control plane adopting a centralized network controller and a direct connection to each data plane node.

Typically, in multi-domain networks, a single PCE is responsible for path computation inside each domain, while the inter-domain path computation is achieved by means of a coordinated communication process among PCEs exploiting the PCEP protocol. Several strategies have been proposed either integrated with the signaling procedure or totally based on the PCEP protocol, exploiting only one PCE deployed in each domain, or building-up a hierarchy of PCEs, as detailed next:

- *Per-domain path computation*, as described in Ref. [37], is a procedure integrated with the RSVP-TE signaling where every time the Path signaling message enters a new network domain the corresponding PCE is interrogated with the PCEP protocol to obtain the path up to the domain border (or the destination if this is the last domain). The procedure is sketched in Fig. 4 and is typically used when transparent lightpaths are terminated at each domain edge node.
- *Inter-PCE path computation*; with this procedure the end-to-end multidomain path is computed before starting the signaling procedure. Specifically, each PCE computation is performed based on the following segment's results that are provided by the following segment's PCE through PCEP. Two versions of this solution have been proposed: the *standard backward path computation* and the *backward recursive PCE-based computation, BRPC* [43,44]. The general procedure is sketched in Fig. 5, showing the PCEP message exchange steps (a-f) among the involved PCEs to compute the end-to-end path and allow inter-domain transparent lightpath provisioning. Optional per-domain PCEP messages are required in the case of inter-operator lightpath provisioning, to compute locally the encrypted path segment provided by the PCEs.
- *PCE-based domain sequence computation*; all the previous methods assume that the sequence of PCE/domains to be traversed is known in advance. Since, in complex multi-domain networks, the selection of the domain sequence may significantly affect the overall network performance, researchers and standardization bodies have worked on defining the procedures combining domain sequence computation and path computation [45,46].
- *Hierarchical PCE solutions*; The (H-PCE) architecture proposal defines the procedures for combining end-to-end path computation with effective domain sequence computation. With H-PCE each domain is equipped with a child PCE (cPCE) for performing intra-domain path computations, while a single parent PCE (pPCE) is responsible for the coordination of inter-domain path computations and computation of the domain sequence. This hierarchical solution provides a further level of centralization that could be effectively exploited to improve network resource utilization [47,48]. The general procedure is sketched in Fig. 6, in which four domains are involved and a light-path is requested between nodes *S* and *D*. The steps *a-d* involve the request to the pPCE (step *a*), the interactions between cPCEs and pPCE with parallel PCReq-PCRep sessions for each involved cPCE (step *b* and *c*) to determine each possible segment computation. The

pPCE computation includes the domain sequence, typically performed using an abstract topology based on star or full mesh of edge nodes, along with the optimal end-to-end path computation, combining the segment computations and replying to the asking cPCE (step *d*). Step *e* shows the final RSVP-based signaling procedure.

The utilization of the aforementioned procedures defined within the PCE architecture might affect the confidentiality of network domain information in a multi-domain multi-carrier network scenario. In Refs. [49,50] mechanisms are proposed to preserve the confidentiality of such information, and possibly discover potential malicious users using the PCEP protocol to retrieve network information. Finally, stateful extensions to the PCE and the PCEP enable the direct LSP instantiation, modification and deactivation request by the PCE to a network node. Novel messages (e.g., PCEP PCInitiate, PCUpdate, PCReport) and objects allow the synchronization of the TED and the LSP control delegation, with the possibility to perform bulk re-optimizations [51].

3. SDN-based control in optical network

Software Defined Networking is a network architecture that enables the management of a telecommunication network, using exclusively software. In this perspective, as shown in Fig. 7 the hardware entities (i.e., network devices), traditionally based on physical and dedicated hardware, are split in two logical segments, representing the physical infrastructure and its configuration (i.e., respectively the data plane and the control plane). The data plane considers all the forwarding components and processes needed to move the traffic along the network. The control plane (i.e., the control logic, represented in red in the figure) is responsible of the software configuration and it is implemented by means of a specific operating system (OS) or an ad-hoc agent (i.e., a software instance) running within the network device.

By configuring and maintaining specific “rules”, the same device (i.e., SDNcompliant generalized hardware) can act as router, switch or firewall, without requiring ad-hoc hardware and showing an advanced flexibility in terms of device administration.

In order to make optimal decisions according to the complete network state, a third entity component is used, called SDN Controller. This element of the SDN architecture communicates with all the network devices, using dedicated interfaces and protocols, being able to both push configuration commands and monitor the state of the instantiated “rules”.

As discussed in Refs. [52,53], the SDN architecture has been applied to optical networks (SDONs). Several industry and research studies have focused on SDON implementation and standardization, with particular interests on the control layer, the infrastructure layer and the application layer. Moreover, SDN paradigm has been also applied to network virtualization, orchestration of multilayer and multidomain networking.



Fig. 8. (a) Partial disaggregation (on the left), (b) full disaggregation (on the right).

This SDN communication interface is used by the SDN Controller in order to send the proper commands to the involved devices, being able to handle the traffic in the expected way. The SDN architecture has been designed for packet networks, but it has been extended in order to be applied to different network segments, including wireless and optical networks. The first solutions considered OpenFlow protocol for the implementation of the control plane interface, since it was the first standard interface proposed by Open Networking Foundation (ONF).

Different works focused on the application of the newest releases of the OpenFlow protocol to optical networks, but, given its native packet-based nature, it presented several limitations, mainly due to the matching rules for identifying the optical traffic, the limited flexibility on the input and output interfaces definition and the simplified representation of the optical layers.

For this reason, other protocols have been considered for the application of SDN architecture to optical networks. Among the proposed solutions, NETCONF represents the de-facto standard. Due to its model-based approach, NETCONF has been successfully applied to the control of optical devices, meeting the requirements and the flexibility needed by optical networks. NETCONF protocol exploits the Remote Procedure Call (RPC) methods, allowing the interaction among the SDN Controller and the SDN agent for both configuration and monitoring. Moreover, custom RPCs can be defined, implementing specific and direct Application Programming Interfaces (APIs).

Initially, each vendor applied the NETCONF-based SDN scheme by proposing proprietary SDN Controller and custom models, exploiting the full configuration of the underlying devices. However, in recent years many activities tried to standardize the SDN architecture by relying on open source projects for the SDN Controller implementation, and the definition of generalized models for the optical devices.

3.1. SDN vs GMPLS optical networks

The first research efforts proposing the utilization of SDN for the control of WSONs are reported in Refs. [54,55]. The work in Ref. [54] demonstrates an OpenFlowbased control plane supporting lightpath setup and release, whereas [55] proposes a possible integration of OpenFlow and GMPLS protocols. Further research work is focused on lightpath provisioning [56,57] with evaluation of optical physical impairments [58,59] in EONs using an SDN control plane.

Those works mainly demonstrated that SDN can be used effectively to manage and dynamically control EONs (independently of the considered protocol enabling the communication between the controller and the node agents, e.g., OpenFlow or NETCONF). However, explicit comparison of the SDN control plane and the GMPLS/PCE control plane is not provided. This kind of analysis is instead reported in Ref. [60], mainly focusing on the provisioning phase. The work in Ref. [41] extends the comparison to the restoration phase, including also an estimation of time needed for node-configuration and path-computation. That work shows that both provisioning and restoration can be significantly improved using a centralized controller performing path computation and parallelized signaling as in the SDN scenario. Moreover, the SDN approach is also able to reduce the setup time with respect to GMPLS-based schemes. Finally, restoration results in Ref. [41] show that under several realistic network scenarios, the SDN approach

reduces also the lightpath recovery time.

3.2. Disaggregated optical networks

The concept of disaggregation, originally introduced in the computing domain, consists in breaking up monolithic systems by separating hardware and software components through open standard interfaces. Disaggregation leads to the introduction of the so-called white boxes (open network systems or elements where different sub-elements can be provided by different manufacturers, even though they are typically operated under the same control system) in contrast to the traditional concept of black boxes (proprietary network systems or elements provided by a single vendor). Disaggregation has recently attracted significant interest in optical networks, since it has the potential to provide cost-effective and vendor-neutral solutions, thus avoiding vendor lock-in. To support the control and management of white boxes, a standard data model is required, so that hardware components and software functionalities can be configured and monitored through common APIs. A key language to describe a standard-defined data model is YANG (Yet Another Next Generation). YANG, standardized by the IETF, has been developed in the context of NETCONF, using XML representation for data encoding [61]. Optical network disaggregation based on NETCONF/YANG has recently progressed to consistent implementations. In particular two main flavours of disaggregation have emerged, called partial and full [62].

The first flavour, called partial disaggregation, is shown in Fig. 8(a). It considers the optical transport infrastructure (e.g., optical line systems (OLS) including Reconfigurable Optical Add-Drop Multiplexers (ROADMs) and amplified links) managed by a single vendor, while transponders, provided in pairs, rely on vendor-neutral NETCONF/YANG control. The most relevant standardization initiative providing YANG models for partial disaggregation is OpenConfig, led by Google.

The second flavour, called full disaggregation, is shown in Fig. 8(b). OLS is further disaggregated, by considering also ROADMs as white boxes to be controlled in a vendor-neutral way. The most relevant standardization initiative providing full disaggregation is OpenROADM, led by AT&T.

So far, partial disaggregation has attracted significant interest from operators and vendors, since it abstracts most of the optical data plane complexity without significantly compromising on transmission performance. Indeed, transponders are provided in pairs, so that transmitter and receiver are from the same vendor, and even proprietary transmission solutions can be implemented by just defining specific operational modes. For these reasons, OpenConfig has gained most of the consensus for transponder control. The full disaggregation flavour appears instead as a more promising approach in the long term. However, the OpenROADM initiative has already provided remarkable standardization results, for example accelerating the definition of standardized pluggable solutions expected to lead to the definition of next generation packet-optical nodes [63].

3.3. Optical network slicing

The network disaggregation is also paving the way toward the possibility to partition a single physical optical network into a number of

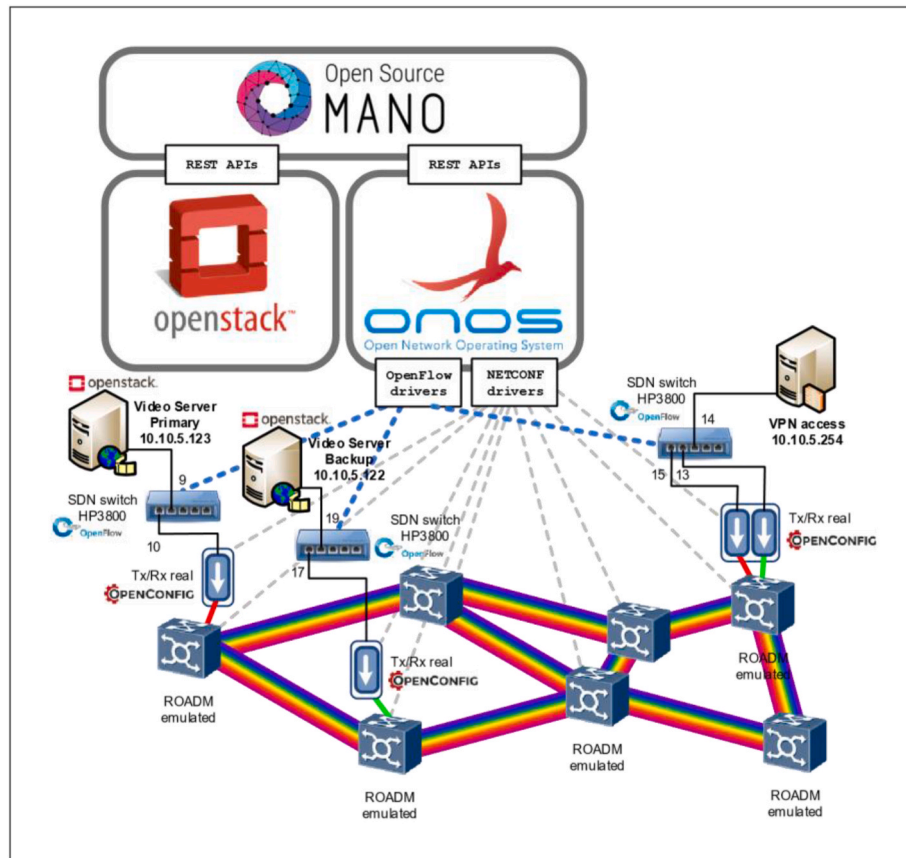


Fig. 9. Multi-layer experimental testbed deploying both a computational resource manager and an SDN network controller.

virtual networks sharing the same physical resources (i.e., network slicing) [64,65].

Specifically, the work in Ref. [64] proposes a network virtualization solution for partially disaggregated networks, based on the concept of a device hypervisor relying on OpenROADM data models. The device hypervisor is implemented by means of an SDN controller utilizing the NETCONF protocol to configure the network devices. The architecture is experimentally validated, considering an EON dataplane provisioning connectivity services across a virtualized network. Starting from a single physical infrastructure, the proposed solution allows the network operator to specify for each created virtual network which nodes and which interfaces should be considered. This solution automatically provides connectivity and performance isolation among network slices because separate physical resources (e.g., different network media channels) are dedicated to each slice. The work in Ref. [65] proposes similar solutions but the performed experimental evaluation is more focused on the slicing of the optical transponders relying on the OpenConfig data models. When the slicing procedure affects packetbased devices, such as the transponder deployed at the edge of the optical network, implementing the slice performance isolation is not straightforward, and some possible techniques deployable in an SDN scenario are proposed in Ref. [66].

4. Multi-layer network control

4.1. Packet-optical networks

The utilization of SDN for the control of the optical layer also facilitates the integration of other network layers. Indeed, the same SDN controller (e.g., the open-source ONOS controller) can be used, through different protocols, to configure both packet-based and optical devices, thus fostering the implementation of networking applications jointly

configuring the two network layers. Moreover, typically well-developed SDN controllers implement also the interfaces towards controllers of computational resources (e.g., OpenStack), facilitating the implementation of an orchestrated framework for the control of both computational and networking resources.

Fig. 9 illustrates an experimental testbed, also demonstrated in Refs. [67,68], including an orchestrator (i.e., Open Source MANO) that communicates with both the ONOS controller (i.e., for networking resources) and the OpenStack controller (i.e., for computational resources). The orchestrator communicates with ONOS to: (1) retrieve network state information (e.g., established lightpaths with utilized paths and optical channels); (2) request the establishment of new lightpaths; (3) request deletion of established lightpaths. The orchestrator communicates with OpenStack in order to instantiate the VNFs.

The considered demonstration provides an example of integrated functionalities required for the dynamic re-configuration of both optical and packet-based networks. Two video servers (i.e., two VMs created by OpenStack) are considered as primary and backup source of a video streaming service. Initially, external users access the video stream from the primary server. Upon failure of the primary server the connectivity has to be re-configured toward the backup server: (i) using the NETCONF protocol the SDN controller activates a new optical lightpath from the VPN access point up to the backup server (the new lightpath is activated on the green wavelength in Fig. 9); (ii) using the OpenFlow protocol the SDN controller re-configures the flow rules at the HP3800 switch located at the VPN access node to dynamically re-write the IP and MAC destination addresses of the video server with the backup server data. This way, the connected clients automatically connect to the backup video stream without any local reconfiguration. All the operations have been demonstrated to be performed without relevant video degradation on the client side.

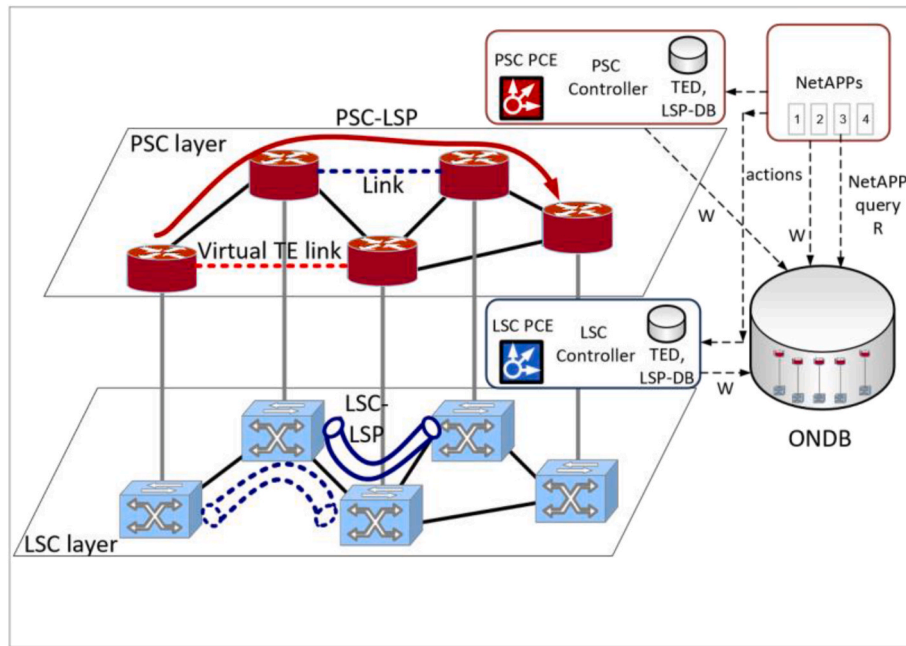


Fig. 10. Multi-layer control architecture including the Open Network Database.

4.2. Multi-domain orchestration

5G and beyond (B5G) radio technologies will enable a new breed of innovative services by the vertical industries (e.g., multimedia, cloud robotics, autonomous vehicles) with more stringent QoS requirements, especially in terms of end-to-end latency [69,70]. In parallel, technological advances enabled by Network Function Virtualization (NFV) and SDN are bringing unique service capabilities to future network infrastructures thanks to (i) elasticity of virtualized network functions deployed in the cloud, and (ii) network programmability enabling flexible network services tailored to the needs of vertical applications (i.e., network slicing) [71,72]. A correlated trend is Mobile Edge Computing, which fosters the deployment of distributed micro-clouds at the network edge (e.g., 5G radio access, Point of Presence (PoP)) to run network functions closer to the users [73].

The resulting network scenario will be a telco cloud infrastructure composed of geographically-distributed clouds (i.e., Data Centers, DCs) interconnected through an all-optical wide area network. DCs span from large cloud DCs located in few remote and consolidated locations beyond the core network (i.e., *core DCs*) to resource-constrained DCs located at the network edge and colocated with telco PoPs in the access segment to offer minimized network latency performance to users (i.e., *edge DCs*). In-between DCs located in the metro/aggregation network segment (i.e., *metro DCs*) complete the infrastructure featuring average processing capacity and processing latency performance. The optical network connects core DCs and arrives up to the access through metro/aggregation links connecting metro DCs and feeding the mobile networks (i.e., 5G radio access) and the network edge (i.e., edge DCs acting as telco PoPs) [74,75].

In this scenario, EONs can play a fundamental role in offering high network capacity, low network latency as well as flexibility features to vertical services, provided that effective resource orchestration strategies are put in place by telco service operators [76]. The foreseen scenario forces a consolidation of the optical and mobile/access network and a convergence of the telecommunication network with the computational resources in the DCs, leading to the challenging task of coordinating the allocations of both network and computational resources (i.e., across different resource domains) to serve virtualized network functions across distributed DCs while meeting (i) computational and bandwidth demands, and (ii) latency requirements of data while traversing and being

processed by DCs (end-to-end latency) [77,78].

In end-to-end latency considerations, two different latency components need to be taken into account along with their inter-play in the considered scenario: (i) network latency experienced by data traversing the DCs inter-connection network links, and (ii) processing latency experienced by data at the DCs servers while being processed. Indeed, in relation to their network locations, the DCs feature not only different computing capacities (and hence different processing latency) but also expose different network latency while data are transferred to other DCs/users, i.e., edge DCs offer higher processing latency due to their lowpower computing systems while being reachable with lower network latency; core DCs offer lower processing latency thanks to their high-performance systems but impose higher network latency due to their distant location [79].

Different cloud and network resource orchestration strategies can be conceived by telco service providers to fulfill network service demands in inter-DC scenarios [80]. The challenge is to address end-to-end latency requirements as well as computational and bandwidth demand while meeting their own management and business targets on, e.g., resource utilization rate, request acceptance rate, risk of Service Level Agreement (SLA) violations [81]. The following criteria can be used for selecting DCs where to deploy virtualized network functions: (i) minimum distance among DCs in order to minimize network latency, although at the cost of higher processing latency due to consolidation of allocations on DCs, (ii) balancing the load among DCs to achieve a fair resource utilization with no minimization in terms of latency, and (iii) minimizing the end-to-end latency at the risk of incurring in higher blocking rate of requests. The strategy to be used is the result of considerations trying to balance opposite targets, i.e. increasing the utilization rate of resources to optimize costs and serve more requests, reducing the risk of SLA violations also due to latency misperformance. Retrial mechanisms in the case of unsuccessful requests can be considered to increase the rate of served requests, although at the cost of increased deployment time [82,83].

4.3. Open network database for multilayer networks

Open-source platforms are considered as network controllers (e.g. OpenDaylight, ONOS). The objective of simplifying the inter-layer workflow without shifting the complexity to a centralized orchestrator

thus preserving a sufficient independence of IP and optical layers has pushed to research an architectural solution based on open-source databases. The goal is to keep the information of the network infrastructure as organized relationships, where software procedures based on multi-match queries simplify several complex procedures such as re-optimization, maintenance, equipment upgrade in an automatic way. The key idea is that keeping the packet-switched and wavelength-switched Traffic Engineering Databases (TEDs) segregated may limit the management and control operation effectiveness [84], while having a shared distributed database with precise reading and writing rights may disclose efficient network operation by means of dedicated network applications.

The scalability performance of modern database technologies are exploited to provide an effective and organized way to correlate TE information originating from different and interconnected network domains/tiers [85]. The TE information defined by YANG of heterogeneous network segments with limited visibility from each other is stored in an Open Network Database (ONDB), that is not enclosed within its PCE or SDN Controller, but it is an independent key element of management and network control architecture, available to any authorized network application which aims to perform defined operations (e.g., maintenance).

Fig. 10 shows the multi-layer control architecture including the ONDB. The ONDB is an additional database with respect to existing databases used to store topology, TED, and installed lightpath information. It includes also information retrieved from a set of existing controller databases and additional information directly stored by the management plane. The stored information models the topology parameters in a protocol independent way, so that it is possible to obtain the path computation even across multiple domains and multi-layer networks. Each network node is defined with attributes that include information related to data plane aspects and configuration information. The ONDB stores the TE information of each layer as provided by the layer controllers. Such controllers have write access to specific relations of the database, whereas the network applications have the read access to the ONDB (even the whole database, if needed). The network applications perform complex search and computations on database relations and as outcome they send a defined list of actions to layer controllers to reach the designated objective.

To assess the effectiveness of such architecture, an accurate analysis on the performance of actual scenarios is needed. In particular, an Optical Link maintenance app has been evaluated in a single-operator multi-layer network. Due to the separation between IP and optical layers management and control maintained by network operators, the Packet Switched Capability (PSC) PCE is not typically able to automatically correlate the optical links used by the PSC LSPs and those stored in the Lambda Switched Capability (LSC) PCE. The ONDB contains layer-specific information that can be accessed by the network application by performing only 3 consecutive simple queries to identify the involved PSC LSPs. The experimental evaluation demonstrated that the proposed architecture and the implementation of the ONDB was able to achieve inter-layer correlation efficiently. The overall optical link maintenance was successfully performed, at the control plane level, in few milliseconds with limited storage resources and impressive scalability performance (i.e., inter-layer maintenance app query time of around 4 ms in a 100 node mesh network with 100 k installed PSC-LSPs).

5. Advanced control and management techniques

5.1. Autonomous optical networks

When deploying optical networks, operators typically adopt conservative approaches in order to maintain the services for their entire life. Thus, worst-case assumptions and margins (as described in Sec. 1.5) are usually considered into QoT figures of merit (e.g., OSNR) to account for QoT estimation modeling inaccuracies [86], traffic-variant physical impairments (e.g., XPM), performance degradation of network devices

at the end of life with respect to the beginning of life [87], and others. Such pessimistic approaches finally result in inefficient network resource utilization and optical reach and channel rate underestimation. For these reasons, in recent years, network operators and vendors investigated the reduction of margins. However, operating a network with reduced margins requires a more frequent monitoring to verify service performance. Consequently, based on the actual physical layer performance, operating services can be re-adapted, e.g. traffic rerouted or transmission parameters such as modulation format, line rate, and forward error correction (FEC) changed [4,5,88]. Such novel approaches are based on the so-called paradigm named “observe-decide-act”, for which decisions (e.g., channel parameters’ values) are taken based on the network observation (monitoring) and then finally acted upon (network reconfiguration). As an alternative to (or in addition to) channel parameter adaptations, algorithms for performance forecasting can be adopted and, as an example, opto-electronic regenerators can be employed at intermediate network life periods when approaching the end-of-life device performance [87].

On one hand, the deployment of coherent receivers – which enables end-to-end performance monitoring through digital signal processing – together with the growth of machine learning in optical networks [89] facilitates the observation of the network. On the other hand, a more frequent collection of monitoring information also results in scalability issues, in particular in an increase of monitoring information to be managed, exchanged, and processed. In a centralized scenario, in case of failures or degradations, the SDN controller, or in general the control module in charge of processing monitoring information (e.g., the Operation Administration and Maintenance Handler [90]), may be requested to handle a huge amount of monitoring information and reconfiguration requests. This can result in delaying service reconfiguration. In order to offer a higher scalability, a hierarchical monitoring architecture has been proposed in Ref. [88].

Each element in the architecture is responsible for a subset of channels and each element can act upon specific types of reconfigurations, thus guaranteeing a shorter reconfiguration delay with respect to a fully centralized scenario [91]. In Ref. [92], another paradigm is introduced to increase scalability: the SDN controller instructs local controllers of network devices on the reconfiguration to apply in case of problems. If a failure or a degradation impacts a service, the SDN controller is bypassed and local controllers promptly react offering a faster recovery/reconfiguration.

Table 3 summarizes possible reconfigurations depending on the type of problem. We have to mention that transmission parameter adaptation may impact the transmitted information rate. As an example, fixing the symbol rate (thus, the switched bandwidth), a lower level modulation format such as PM-QPSK offers more robustness than a higher level format such as PM-16QAM, but it reduces the bit rate (by two times in the case of PM-QPSK and PM-16QAM). Similarly, keeping fixed the symbol rate and introducing more redundancy (FEC adaptation) results in an information rate decrease. Node or link failure implies a complete loss of connection, thus, in this case, traffic should be re-routed: in case of protection, the traffic is simply switched to the protection path, while in case of not protected traffic, the SDN controller should be invoked for dynamic restoration. Performance degradation (e.g., due to amplifier ageing) implies a change on the QoT figures of merit: e.g., an increase of the bit error rate (BER) or a decrease of the OSNR. Performance degradation may be solved with the change of the modulation format or

Table 3
Failures/Degradations and reconfigurations.

Failure/Degradation	Impact	Reconfiguration
Node or link failure	Loss of connection	Switch to protection path or dynamic restoration
Device performance degradation (e.g., amplifier)	BER increase or OSNR decrease	Switch to protection path or FEC/Format adaptation plus dynamic restoration

the increase of the redundancy (Forward Error Correction - FEC). As an example, with the same symbol rate, PM-QPSK requires about 6.5 dB-less OSNR than PM-16QAM [93]. As stated before, when passing from PM-16QAM to PM-QPSK or when adapting FEC, the bit rate is reduced: in this case, part of the traffic can be promptly recovered with transmission adaptation, while the rest should rely on the SDN controller for dynamic restoration [93].

Finally, performance degradation due to traffic-variant impairments such as XPM could be overcome with dynamic reconfigurations such as rerouting or the shift in frequency of the channels (indeed, the impact of XPM decreases with the distance between interfering channels in the spectrum [94]). However, we believe that the reduction of margins due to XPM is challenging. Indeed, QoT estimation based on the actual traffic instead of the worst-case scenario implies that, when establishing a new channel, the QoT of already deployed channels is re-estimated/re-observed. This may imply intensive computations and operational procedures for a network operator. Thus, the QoT estimation based on the XPM worst-case assumption may still remain the most practical solution: if a channel is acceptable in the worst case, it is acceptable with any traffic deployment and no actions are requested due to XPM.

5.2. P4-based programmable data plane

To address TE, packet-layer substantial innovations such as Segment Routing [95] also enabling SDN/Network Function Virtualization (NFV) Service Chaining [96] and Network Slicing [97] have been proposed to automatically steer per-service traffic slices onto multi-layer EON light-paths. However, such procedures rely on flows with bandwidth reserved in the control plane only. In the data plane, short-life, bursty and unpredictable behavior of traffic forwarded by IoT gateways, 5G fronthaul, metro aggregation nodes may be subject to different time-dependent profiles and statistical variations, that may induce bottlenecks, congestion, queue delay, thus affecting QoS. This is particularly challenging for time-sensitive traffic. Indeed, traditional SDN implementations do not support such required stateful-driven forwarding at wire speeds directly at the nodes, whereas the controller is typically involved to react upon critical events. However, this may pose serious scalability issues at the controller and may noticeably delay reaction enforcement at the data plane, thus leading to serious forwarding inefficiencies (e.g., unexpected latency and jitter increase). In addition, at the data plane level, dedicated and vendor-locked fixed-function hardware represents the main limitation affecting SDN flexibility and configurability.

Current data plane technologies for networking, including multi-layer, are drastically evolving in the last years in terms of programmability. One of the main candidate technological drivers for such evolution is the P4 programming language.

The P4 (Programming Protocol-independent Packet Processors, p4.org) is a high-level programming language explicitly devoted to the packet processors data plane. P4 has been conceived in the SDN paradigm. The novelty and the potentials introduced by P4 have gained significant attention and interest by many chip manufacturers. Today, the P4 ecosystem consortium includes more than 50 industrial corporations (e.g., Cisco, Dell, Google, HPE, Intel, Huawei, Juniper, Microsoft, ZTE) and many academic partners. The main features of the P4 language are the following: 1. Independency from the target hardware specification; 2. Independency from any specific network protocol; 3. Reconfigurability on the field, redefining the packet parsing and processing rules when needed.

Compared to the state-of-the-art of packet processing systems, based on micro-code, P4 provides the configurable building blocks of an abstract network node, ranging from Layer 0 up to Layer 7 functions: parsers (including non-standard headers), controls, tables, actions. In addition, packet-forwarding policies, algorithms and per-packet custom actions can be implemented, producing portable implementations over different hardware targets (e.g., network interface cards, FPGAs,

software switches and hardware Application Specific Integrated Circuits - ASICs). Definitely, P4 does not represent an extension or an evolution of the OpenFlow protocol, since SDN control plane protocols do not really control the internal behavior of a switch, being the control plane kept separated from the forwarding plane. Conversely, the OpenFlow protocol can be seen and implemented as a P4 program. Indeed, P4 allows the custom data plane programming of any switch, regardless of how it is controlled, thus widening the scope to scenarios or use cases where SDN control may not be suitable (e.g., industrial switches with internal operating system). P4 compilers are able, besides the generation of an executable switch, to define and auto-generate the API needed to populate the node table entries.

P4 presents two features that are of relevant interest with respect to traffic engineering. First, P4 allows stateful programming objects, such as counters, meters and registers, enabling context-aware decisions. Second, the dimension of the hardware resources needed for packet processing and stateful objects can be explicitly defined directly in the P4 program, thus allowing dedicated (and even dynamic) resource allocation. Third, P4 programmability includes custom headers and packet metadata. This opens the way to three novel features: 1) the possibility to dedicate pre-planned and dynamic countable hardware resources to specific functions and processes; 2) the possibility to extract features and store states and packets headers or fields directly in the P4 node for pre-processing purposes and threat analysis; and 3) the possibility to customize protocol stacks, to realize dedicated functions and protocols between the P4 switches (e.g., inband telemetry). Applications of P4 are numerous for implementing advanced packet-layer functions in a multi-layer node: stateful traffic engineering for tributary interfaces [98], in-network awareness at the edge [99], multi-layer network telemetry [100], cybersecurity functions [101], 5G User Plane Function offloading [102]. Moreover, in the latest node architectures, optical pluggables will be envisioned to be controlled directly inside a programmable packet-switched backplane running open operating systems (e.g., SONiC) and leveraging the SDN P4 programmability [103]. As shown in Fig. 11, the P4Runtime API allows the SDN controller to push a specific P4 code to a programmable node, thus instantiating the programmable pipeline functions, actions and controls. Moreover, it allows the discovery of the nodes and their flow entries taxonomy, enabling the direct control of the node by sending the flow entries to configure the behavior of each P4 function/application and the specific flows to be subject to P4 advanced treatment. The degree of freedom of such architecture represents an unprecedented level of configurability and tunability, that opens the way towards a fully programmable and autonomic SDN network.

5.3. Intent-based networking

The growing market of network services requires high dynamicity and flexibility of the network to address requirements of emerging applications (e.g., telemedicine, hyper-connected smart cities, industrial automation) enabled by 5G (and beyond) capabilities. The main reference architectures, based on the innovative paradigms (i.e., NFV, SDN and orchestration) are presented in Ref. [104], highlighting the impact on Operations Support System/Business Support System (OSS/BSS), while analyzing the open challenges of this topic.

The success of emerging applications and future network operator's revenues will depend on the ability of network operators to effectively address heterogeneous network resource demands and wide dynamics originating from a highly changing context of users, data and services. In this regard, the adoption by network operators of some degree of automation and orchestration [105,106] coupled with novel service models [107,108] and application-to-network interaction paradigms [109] will be relevant.

In order to support efficient and faster service delivery as required by emerging application services, the Intent-based Networking (IBN) concept represents a new approach to network management where a

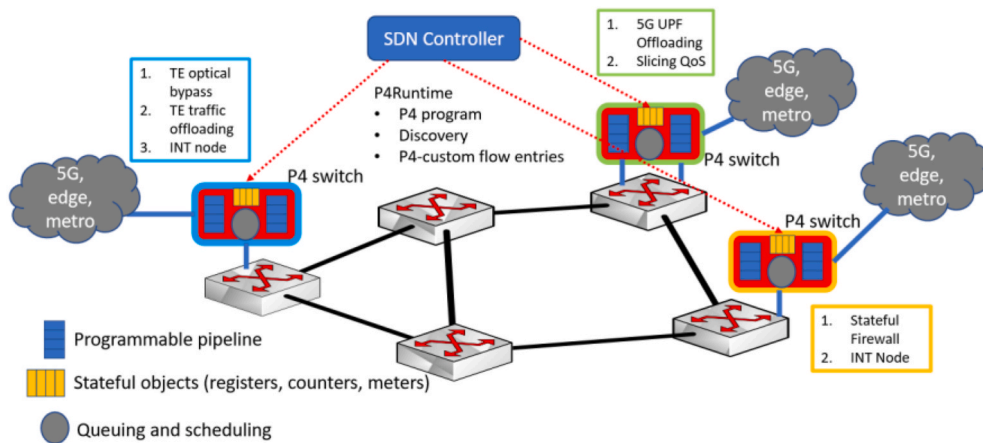


Fig. 11. Programmable SDN P4 switches as advanced packet-layer nodes.

specialized software layer can (i) simplify network management and operations, (ii) automate the implementation of network configurations required by applications, and (iii) ensure their full lifecycle management in line with expressed goals.

The IETF has started promoting the IBN as a networking model that is based on a declarative approach. In IBN, users at an application layer (e.g., customer's OSS/BSS) or a service management layer (e.g., network operator OSS/BSS) can express their business, service or operational goals through highlevel prescriptive directives (i.e., Intents), thus unburdening applications from the technology-specific low-level networking directives needed for those goals. Examples of intents are: "set a connection as a high availability network service", "always maintain high quality of service and high bandwidth for gold level users", "provide connection C with guaranteed bandwidth for customer A". On the other hand, the network operators are left with the flexibility of addressing the expressed goals based on their own optimization decisions, in light of a loosely-coupled interworking with applications [110,111].

An IBN system (i.e., *Intent layer*) translates the high level declarative goals into actions (including configurations) that are automatically deployed within the network and maintained by a set of management function blocks in one or more network management/control systems. Indeed, not only configuration aspects are covered. Also the continuous tuning of network state (derived through real-time or periodic measurements) is addressed to adjust it to the expected state based on the declared business goals. More specifically, the Intent Layer manages and regulates the lifecycle of Intent demands from applications through *fulfillment* and *assurance* operations in a closed loop workflow. Hence, IBN not only contributes to increase automation but also to improve the robustness of the network through closed loop and to reduce costs through dynamic network operation and maintenance.

More specifically, the Intent layer includes five main functionalities: (i) the *Translation*, which parses the application's Intent and translates it into a set of networking actions, mainly configurations; (ii) the *Management and Decision*, which are responsible for identifying the needed actions to ensure that the intent is achieved including derivation of the algorithm to be used, learning on how to optimize outcomes over time, and coordination of configurations and deployment actions, e.g., rendering of high-level abstractions into lower-layer parameters, (iii) the *Analyses/Verification*, which continuously verifies the status of the Intent, and if necessary triggers corrective actions leveraging on Management and Decision block, and finally (iv) the *Intent Repository*, which is a database that interacts with the intent Management and Translation modules to provide mapping between the "intent" and its "configuration" [111].

IBN attracted increasing interest as an evolved way to manage networks. Currently, researchers from standardization organizations [111, 112], open source communities [113,114], industry [115,116] and academia [117] are actively investigating the mechanisms and the

applications of IBN in different fields of networking. Traditionally, research in IBN focused on core networks [118]. Recently, IBN embraced also other networking fields, such as mobile networks [109] as well as network slicing as part of slice management functions [119]. As for optical networking, noteworthy research works include [120, 121]. Finally, an IBN approach coupled with the use of learning techniques is considered promising especially to improve the (i) translation process using an AI agent trained through natural language processing techniques, (ii) optimization of the intent rendering or intent refinement process, (iii) anticipation or forecasting of the future system state and response to the changes in intents or network conditions. Further details can be found in Ref. [122].

6. Recent research challenges

6.1. Zero-touch network and service management in optical networks

The Zero-touch Network and Service Management (ZSM), proposed by ETSI [123], is among the emerging architectures for autonomously operating and managing communications networks. Research papers and projects already reported studies related to network management automation [124]. For example, Deep Learning (DL) applications for various traffic control aspects, such as network traffic classification, network flow prediction, mobility prediction, Cognitive Radio Networks (CRNs), and self-organized networks have been proposed in Ref. [125]. A closed-loop solution for network slicing, where traffic forecasting information is ingested by an admission control engine to maximize the number of granted network slice requests while meeting the slice Service Level Agreements (SLAs) guarantees, is presented in Ref. [126]. Forecasting/predictive analytics has been proposed in Ref. [127] for scaling 5G core network resources by anticipating traffic load changes. The approach is based on two Artificial Intelligence/Machine Learning (AI/ML) techniques: Recursive Neural Networks (RNNs), more specifically Long Short-Term Memory (LSTM), and Deep Neural Networks (DNNs). The results show that forecast-based scalability mechanisms outperform threshold based solutions in terms of latency to react to traffic change. In Ref. [128] a first demonstration of ROADM White Box augmented with machine learning capabilities is demonstrated. The white box includes various levels of disaggregation, NETCONF/YANG control, telemetry and spectrum-based advanced monitoring functionalities.

However, in all the aforementioned papers two assumptions are typically considered: all the control loop functions are co-located or the delay in exchanging data among the control loop functional elements is not taken into account. One of the first studies taking into account the delay introduced by the data collection is presented in Ref. [129]. There, the time response of a Hybrid Forecast Framework (HFF) running at the

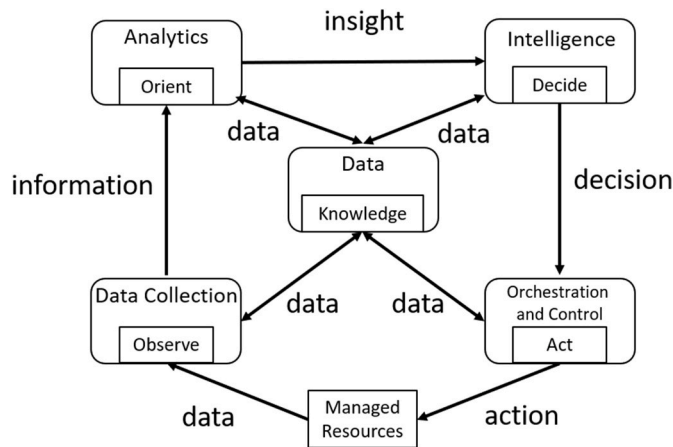


Fig. 12. Mapping between ZSM architecture building blocks and closed loop functions as in [123].

network edge or in the cloud is compared.

When running at the edge, the HFF exploits traditional time series analysis prediction algorithms to minimize the utilized resources and energy while it exploits AI/ML tools to make predictions in the cloud. Results show that if the look-ahead time is long, cloud-based prediction is preferable because the long look ahead time compensates for the higher time for prediction, mainly due to data transfer.

In [130] the authors discuss on the need of innovative service and resource orchestration platforms, based on existing approaches and combining centralized and distributed approaches on heterogeneous infrastructure. They focus on the network control as the core component, presenting multidomain and multilayer networks use cases, with particular interest on joint cloud/network orchestration and 5G network slicing.

Other architectures than the ZSM are proposed by standard developing organizations: the TM Forums Open Digital Framework (ODF) [131], MEF 3.0

Lifecycle Services Orchestration (LSO) [132], Linux Foundation Platform for Network Data Analytics (PNDA) [133], TM Forum Smart (BPM) [134]. Even research projects are exploring solutions that provide management automation. SLICENET [135] is a 5G-PPP phase II project that aims at designing and implementing an end-to-end cognitive vertical-oriented 5G network slicing framework. 5G-enabled Growth in Vertical Industries (5 Growth) [136] aims at empowering vertical industries such as Industry 4.0, Transportation, and Energy with Artificial Intelligence-driven automated 5G end-to-end solutions that allow them to simultaneously achieve their respective key performance targets.

In general, the ZSM shall provide means for the ordered invocation of the phases of a closed-loop control. An example is the closed-loop devised by J.R. Boyd [137], also known as the OODA loop: the Observe, Orient, Decide, and Act. Thus, it is possible to find a mapping between management functions and closed-loop steps: Data Collection contributes to Observe, Analytics contributes to Orient, Intelligence contributes to Decide, and Orchestration and Control contribute to the Act steps as shown in Fig. 12.

In the classical control engineering, the performance of a system that is controlled by a closed-loop depends on many factors, such as the feedback that is applied. Typical key performance indicators are the system time response and the stability. In ZSM, due to the complexity of the utilized functional elements, the performance depends not only on the time it takes to execute each functional element of the control loop but also on the time it takes to exchange data among the functional elements. Such times impact the reaction time of the system, that can be assimilated to the system response time of the classical control engineering closed-loop.

6.2. Telemetry-based optical network control

In recent years, the management and control plane of optical networks has rapidly evolved, moving the attention to the real-time analysis of key transmission parameters [138]. In fact, by taking in consideration the main parameters of the optical infrastructure it is possible to improve network reliability by speeding up failure detection, failure localization, and so the overall recovery processes, particularly in case of soft failure, where only few parameters have limited perturbations [139].

The possibility to efficiently observe the real-time behavior of the key parameters of transponders, amplifiers and Reconfigurable Optical Add Drop Multiplexers (ROADMs) becomes crucial in the context of the SDN paradigm, allowing the SDN controller to take optimal decisions and avoid possible SLA violations.

In traditional architectures, optical network devices were managed using proprietary APIs (i.e., Local Craft Terminal applications), exposing monitoring functionalities for the main parameters of amplifiers and transponders with the minimum granularity of 15 min/1 h, in order to not overload the control plane network. By relying on standard monitoring protocols (i.e., Simple Network Management Protocol - SNMP or Hypertext Transfer Protocol - HTTP), such monitoring solutions allowed the periodic polling of the network devices, enabling the collection of data with limited data rate and relaxed time interval, while generating asynchronous messages in case of specific events (i.e., link failure or parameter exceeding a threshold).

New generation monitoring functionalities have been designed in order to provide the so-called telemetry services, allowing the fine retrieving of real-time values by extending the SDN agents and the underlying hardware drivers and enabling on demand streaming of the selected transmission parameters. In fact, thanks to the control plane high bandwidth availability, innovative and effective monitoring solutions have been proposed for open devices and agents, enabling the massive collection of real-time parameters at the sub-second sampling times [140] to be analysed by artificial intelligence (AI) and machine learning (ML) analytics functional blocks, able to process such data, activating automatic and proactive SDN routines. Typically, the telemetry is activated on the network device where the telemetry server is available, and a data streaming is enabled towards a telemetry collector, including a variable number of parameters, according to the needs of the SDN Controller. For this purpose, novel protocols (i.e., generalized Remote Procedure Call, gRPC) and data modeling (i.e., protobuf) have been adopted, providing the new generation monitoring systems with high flexibility and custom configurability, while maintaining communication efficiency [100,141]. Recently, telemetry architectures and workflow solutions have been proposed for the following use cases: multi-layer networks encompassing packet-switched P4 programmability [100], decentralized reoptimization at next-generation optical pluggables [142], adaptation of existing monitoring tools such as Kafka in the disaggregated optical framework [143].

6.3. Optical packet switching in data center networks

For the data center connectivity, tailored solutions are required. In data centers, the interconnection network can be partitioned in different segments, as shown in Fig. 13: interconnection of the servers within the same rack (intra-rack), and different racks (inter-rack). For large scale data centers, an additional level of interconnectivity may also exist, consisting of a subset of racks interconnected to form a cluster.

For the *inter-rack segment*, many different types of packet-switched optical networks have been proposed to be used within data centers. Optical packet-switched rings is one of the most suitable solutions that have been well investigated in various projects, including DBORN [144], DAVID [145], HORNET [146], FLAMINGO [147], RINGO [148], HOPSMAN [149], and ECOFRAME [150]. An alternative solution is the Torus architecture [151] and Clos architecture, e.g., in LIGHTNESS

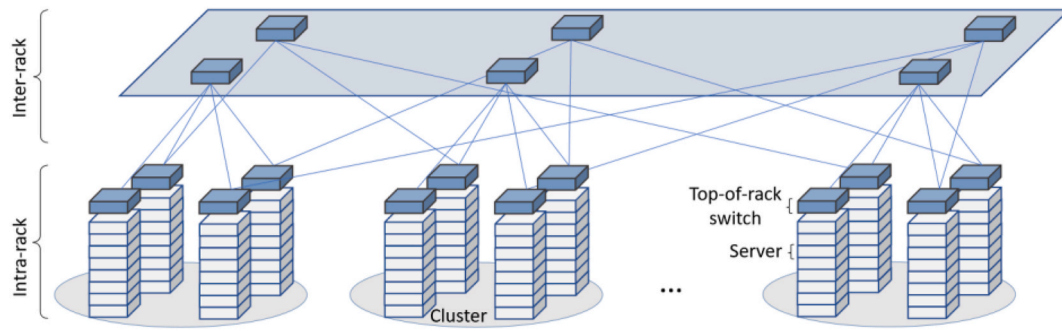


Fig. 13. Example of data center network.

[152] and ROTOS [153] projects and in Ref. [154]. Wavelength-based addressing has been considered in switch architectures based on arrayed waveguide grating routers (AWGRs) and tunable lasers or wavelength converters [155–157].

To properly control the data packets injected in the intra-rack segment network, optical packet schedulers are necessary when the resources (e.g., wavelengths) are shared among different connections, that is the optical slots of each wavelength can be assigned to different connections. The scheduling problem can be solved in a centralized manner (e.g., using permutation matrix [158]), with grant and arbitration logic [154], or in a distributed manner [159]. Scheduling may need to account for resource limitations and constraints, and QoS (e.g., in terms of data rate and latency) [159]. Demonstrators of controllers have been implemented in FPGA [152,158,159] and ASIC [154]. Additional features include QoS [159], slice ability [159] and SDN control [152,158,159] along with switching capabilities from microseconds [159] up to nanoseconds switching time with a centralized controller [154].

For the *intra-rack segment* interconnecting the servers, typical solutions exploit a single optical domain (e.g., wavelength) to enable switching and avoiding contentions. For instance the wavelength domain is exploited for switching in OSMOSIS [160] and OPSSquare [161]. The interconnection architectures can be based on Clos, crossbar or full mesh topologies. A more scalable architecture is the multi-plane one, which exploits two switching domains, such as space and wavelength, space and time, orbital angular momentum (OAM) and wavelength [162–164]. The use of two switching domains allows the realization of hierarchical switching architectures, where a first switching domain addresses switch cards, and a second switching domain addresses ports within each card, which eases the physical impairment and scalability issues typical of optical solutions.

All these architectures for the intra-rack segment require a fast and scalable scheduler and controller, which is typically implemented in a centralized manner. For single-plane switching architectures, conventional scheduling algorithm (e.g., based on maximal matching algorithm) can be used, as well as parallel approaches trading performance for computational speed [165]. For multi-plane switching architectures, a tailored scheduling algorithm is required, such as the two-step approach presented in Ref. [166], which parallelizes the calculations in the two domains, enabling a fast and high-performance scheduling. Implementation in FPGA and ASIC have been realized for the intra-rack schedulers and controllers, demonstrating scalability up to 1000 ports in a ASIC implementation [165] and a time for computation and switching in the order of microseconds [164,165,167].

Finally, the integration of the many interconnection network components into a photonic integrated circuit (PIC) is nowadays achievable with current integration platforms, as recently surveyed in Ref. [168]. However, the photonic integration of larger and faster switching architectures opens new challenges in terms of electro-optical packaging and electronic switch control [168,169].

7. Conclusions

This survey presented a historical perspective, the state-of-the-art and the major research efforts carried out in the recent years in the context of management and control of optical networks, with specific focus on metro-core and data center networks.

Interestingly, the management/control architectures followed a circular evolution bouncing between centralized and distributed architectures along years. In fact, early optical network configuration has been based for a long time on a centralized approach adopting exclusively the Management Plane. When dynamicity called for an automatic set-up and tear down of lightpaths, the first widely deployed model of control plane was distributed and based on the GMPLS protocol suite. Such an approach, still responding to the aforementioned needs, exhibited some shortcomings in connection with the concurrent set-up during provisioning, restoration and uncoordinated operation. Hence, the centralized control was partially re-introduced through the use of the PCE. The PCE has been described in terms of architecture, single-domain, multi-layer, multi-domain operation, also covering innovative database update synchronization and optical impairment-aware path computation. The centralized approach in network control was fully revitalized when SDN-based control was introduced with full relevant features, enabling a novel dimension of interoperability of optical networks and giving rise to the so-called “disaggregation”. The following advanced topics are also presented in the survey, covering the issue of automatic optical network operation: autonomic optical networks, where through a set of instructions network nodes are able to react on network changes; the use of P4 programming language to condition certain classes of traffic at the data plane level and intent based networking, allowing a high-level control of optical networks.

Most recent approaches still in the research phase are also detailed, namely the zero-touch network and service management in optical networks (based on a European Telecommunications Standards Institute - ETSI recommendation), the use of telemetry for network monitoring and the optical packet switching networks in data center networks. All sections are supported by consolidated results provided by large amount of studies in the literature. As thoroughly reported and elaborated in the paper highlighting, optical network management and control have been and will be a lively research topic for the next decades. It is also worth to stress that time-sensitive networking is a trend that is becoming quite stringent for metro-core networks too, advocating on the long term some of the intra-data center technologies on a wider geographic scale.

Finally, this survey is intended to present an overview of the main streamline research in the area according to the authors' view and it is meant to be offered to the scientific community working on novel management and control solutions for optical networks.

List of acronyms

The list of acronyms is provided in Table 4.

Table 4
Acronym list.

Acronym	Description
16QAM	16 Quadrature Amplitude Modulation
AI	Artificial Intelligence
API	Application Programming Interface
ASE	Amplified Spontaneous Emission
ASIC	Application Specific Integrated Circuit
ASON	Automatic Switched Optical Network
AWGR	Arrayed Waveguide Grating Routers
B5G	Beyond 5G
BER	Bit Error Rate
BGP	Border Gateway Protocol
BRPC	Backward Recursive PCE-based Computation
BSS	Business Support System
BV-WSS	Bandwidth Variable Wavelength Selective Switch
cPCE	child Path Computation Element
CRN	Cognitive Radio Networks
DC	Data Center
DL	Deep Learning
EON	Elastic Optical Networks
ETSI	European Telecommunications Standards Institute
FCAPS	Fault, Configuration, Accounting, Performance and Security
FEC	Forward Error Correction
FPGA	Field Programmable Gate Array
GbE	Gigabit Ethernet
G-LSP	Generalized Label Switched Path
GMPLS	Generalized MultiProtocol Label Switching
gRPC	generalized Remote Procedure Call
HFF	Hybrid Forecast Framework
H-PCE	Hierarchical Path Computation Element
HTTP	Hypertext Transfer Protocol
IBN	Intent Based Networking
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
IP	Internet Protocol
ITU-T	International Telecommunication Union - Telecommunication
LMP	Link Management Protocol
LSC	Lambda Switched Capability
LSO	Lifecycle Services Orchestration
LSP	Label Switched Path
LSTM	Long Short-Term Memory
ML	Machine Learning
NFV	Network Function Virtualization
OAM	Orbital Angular Momentum
OLS	Optical Line Systems
ONDB	Open Network Database
ONF	Open Networking Foundation
ONOS	Open Network Operating System
OODA	Observe, Orient, Decide, and Act
OSNR	Optical Signal to Noise Ratio
OSPF-TE	Open Shortest Path First with Traffic Engineering
OSS	Operations Support System
OXC	Optical Cross Connect
P4	Programming Protocol-Independent Packet Processors
PCE	Path Computation Element
PCE	Path Computation Element Protocol
PIC	Photonic Integrated Circuit
PMD	Polarization Mode Dispersion
PM-QPSK	Polarization Multiplexed Quadrature Phase Shift Keying
PNDA	Linux Foundation Platform for Network Data Analytics
PoP	Point of Presence
pPCE	parent Path Computation Element
PSC	Packet Switched Capability
QoT	Quality of Transmission
ROADM	Reconfigurable Optical Add-Drop Multiplexer
RPC	Remote Procedure Call
RSA	Routing and Spectrum Assignment
RSVP-TE	Resource Reservation Protocol with Traffic Engineering
RWA	Routing and Wavelength Assignment
SDH	Synchronous Digital Hierarchy
SDN	Software Defined Network
SDON	Software Defined Optical Network
SLA	Service Level Agreement
SNMP	Simple Network Management Protocol
SONET	Synchronous Optical Network
SONiC	Software for Open Networking in the Cloud

Table 4 (continued)

Acronym	Description
SPM	Self-phase modulation
TED	Traffic Engineering Database
VNF	Virtual Network Functions
WDM	wavelength-division multiplexing
WSON	Wavelength Switched Optical Network
WSS	Wavelength Selective Switch
XPM	Cross Phase Modulation
YANG	Yet Another Next Generation
ZSM	Zero-touch Network and Service Management

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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